Final Model Description

The model used for prediction is a random decision tree forest that takes in most available factors to predict the log final price. The features used in this model include:

From train.csv:

- Dummy variables for series and artists with more than 10 entries in train.csv.
- Dummy variables for different common types of damage.
- Original.
- Length of title.
- If the item was on clearance.
- If the item had a notably low starting bid.
- A binary indicator for whether a handful of other words are found in the condition, such as “Panel” and “no issue”.

From outside train.csv:

- The number of times an artist appears in a 2021 dataset from ukiyo-e.org.
- The era an artist or artwork is from, built from a 2021 dataset from ukiyo-e.org.

These variables go through a random forest with 800 trees, a minimum samples per leaf of 5, and max features of 0.5.

The Python script will output two things: the training data with predictions and the testing data with predictions.

Run Instructions

These instructions are for the Mac terminal, which are the ones I used.

- Get the environment ready with:
    - Change directory (cd) into the folder containing scripts, test.csv, train.csv, and raw data.
    - python3 -m venv venv
    - source venv/bin/activate
    - pip install pandas numpy scikit-learn
        - Or: python -m pip install numpy pandas scikit-learn (which I had to use because pip was not found)

- Transform the test and train data using the two scripts; these output two new .csv files with the specified data from data_raw.csv included, you may need to change the paths in the python file for this.:
    - python3 transform_test.py
    - python3 transform_train.py
- Now you can run the random forest:
    - python3 rf_final_bid_final.py
- This will output test_with_predictions.csv and train_with_predictions.csv.

How I Went About Making This Model

Notes:

This was difficult at first, but some realizations came to me. I also readily admit that better models could be made with tools discussed in class. This attempt uses tools whose methodology I understand best, as I find it irresponsible and wrong to turn in something I do not understand. However, I believe the model I have made, at least going off of the data produced by it and its validation, has proven itself sufficient even if simple. I have learned a lot from this project on how to use Python, but there is a lot more that can be done with more training.

AI has been used to write much of the code, as allowed by the rules set out in the rubric. This was primarily for three reasons:

A. I am inexperienced with string manipulation without assistance.
B. I am inexperienced in Python compared to R.
C. It helped me use and learn the scikit-learn package.

This does not mean I do not understand each step of the code, as I hope my many comments point out. Rather, I am less familiar with the syntax of Python and routinely make mistakes on my own.

AI was also used to attempt to make better models, which I will discuss later. I had Claude write an R script that was able to predict the data with an R^2 of ~0.85 using XGBoost and CatBoost. However, I do not understand how these models specifically work, nor do I understand the code written, so it would be dishonest of me to use this in my final output.

Steps to Build the Model

The beginning of this project consisted of attempts at visual and linear regression modeling in R. This came in the form of examining plots, subsetting data, and visually looking

for trends in the data. At points this became futile, as relationships that seemed obvious to me at first showed little to no relation after research; these include:

- Older works being more valuable.
- Famous artists being more valuable.
- Different types of damage, such as staining or creases, having a negative effect on average price.
- How late in the artist's life a work was made having an effect on price.
- The description of the piece as “nice”, “excellent”, or “good” being treatable as actual descriptors of quality and therefore price.

However, some inferences, like originals being valued more than non-originals, were clear.

Looking back, I could have come out of this slump much faster if I had:

1. Looked at the raw data more closely first.
2. Understood the auction process on the site.
3. Done preliminary research into ukiyo-e appraisal.

These three were major revelations that helped me work on my dataset and recognize the appropriate model.

My realization that sparked the primary method to be used, decision trees, was caused by scrolling through the actual dataset. Here, I discovered that the title and series would often note that a piece was on sale or had a low starting bid price. When checked, this was a major indicator of price, as many pieces on clearance would go for similar low prices. However, it also did not fully explain price, as even pieces with low bid prices would sometimes reach the $200s if by a famous artist. From here, I learned I would need a model that would split the data depending on certain categorical variables and explain relationships that would be impossible to see without trial and error. This clearly called for a decision tree.

The first attempts proved marginally more fruitful than before, as even with just a decision tree predicting based off different types of sale and the year of the piece, I was getting moderately correlated predictions.

I had also found that many entries are duplicates and follow a certain pattern: a piece will be listed with a starting bid; if it is not purchased, a week later the starting bid drops by $5-$10 or stays the same. This process repeats an indeterminate number of times until the item is bought, placed on sale, or removed.

One extremely useful source for how Ukiyo-e is priced was this video Woodblock Print Value

This pointed to a handful of factors that I should attempt to address in any model:

- While artists such as Kuniyoshi, Yoshitoshi, and Kunisada from the “golden age” of ukiyo-e are well known and praised, the reproducible nature of woodblock printing meant

many of their works were heavily printed. This explains why famous artists did not consistently command high prices; rather, they exhibited high variance in price.
- Despite the fact above, artists play an extremely important role in predicting price, as high-value art is still often made by them.
- There are many well-known and highly valued series, so series (not including sales listed as series in the data) need to be accounted for in the data.
- Edition is an important factor, but one that cannot be readily identified within the bounds of my ability. Even though the tools needed are allowed for this project, I do not have the ability to use them properly and have decided to forgo them.
- Condition has a small role to play in evaluation, but it can often be overshadowed by artist, edition, and series.

Yet, even with all of this, the decision trees that were formed usually stopped giving useful branches after checking the artist and whether the piece was on sale. At this point, I reached another stopping point. While I knew what should predict price, a standard decision tree wasn't cutting it.

At this point, I decided to have an AI attempt to make a model. Claude gave code for an XGBoost random forest—a tool mentioned in class, but one I didn't understand well enough to feel comfortable using. This gave an R^2 of ~0.85 for the relationship between predicted and real prices, a very high number. This led me to attempt a simpler version using a basic random forest, whose methodology I understood. One thing it still struggled with was artworks with values of ~$700+, consistently undercounting them. Combined with the fact that I found no conclusive similarity between most high-value pieces, this tells me it may not be good to focus too much on these, even if they produce high errors.

Here, I switched from R to Python. With the help of Python to manipulate strings and use the scikit-learn package, I was able to get a fairly good result. For this model, I also properly implemented dummy variables for artists and series.

“1.0 Model”:
This is a random forest that took in these variables:
- Boolean indicator for whether the condition contained any reference to various words indicating damage.
- Dummy variables for artist and series, marking any artist or series with fewer than 10 entries as "Other Artist" or "No Series", respectively.
- The year listed in some piece titles.
- If clearance is in the string for series or artist (done before artist was converted into a dummy variable).
- If various words indicating low starting offers were present in the series column.

This worked remarkably well, with clearance listed as by far the most important feature.

This would then use out-of-bag predictions to predict prices for the training data, which I used to check how well the model predicts.

This was further validated with a simple 10-fold cross-validation to test consistency. This gave a relatively similar RMSE, indicating that the model may not be overfitting or a fluke.

When looking at the plot of log real vs. predicted prices and the output table, I saw that the model struggled with some large series by a single artist. These showed up as large vertical streaks in a graph of log predicted price vs. log final bid. This was an issue I felt could easily be tackled.

After this general success, some tweaks were in order:

- Year in title was included. While shown to be an important feature, in the long term I knew this was controlling for a collider in the artist-price relationship, which is bad for prediction.
- I added separate variables for items mentioned in condition that I was unsure would raise or lower price, such as: "no issues", "backing", "signature", and "panels".
- I also added title length. This was the most important variable in the AI's model, and I believe it has alot to do with duplicate entries, but I'm not entirely sure.

After another test, all of these improved the model except for signatures. This also removed the earlier issue where large series were predicted at the same price.

Words marking damage were present over 70% of the time, so this was split into dummy variables for specific damage types, as well as some uncommon ones.

Various other tweaks were made to the model, mostly removing irrelevant features, fixing errors, and making code cleaner.

Additional Sourcing

Because the dataset given was limited, I used statistics derived from https://zenodo.org/records/13120879. This dataset describes most artworks found on ukiyo-e.org in 2021, a repository of images and descriptions of Japanese woodblock prints. Two pieces of data I extracted to improve this model are: a count of how many pieces were created by the artist, and the era the artist belongs to.

Artists appearing fewer than X times were grouped into an "Other" category, as infrequent artists offer too little data for decision tree splits. Era serves as a broad category to distinguish famous historical periods in ukiyo-e. Artist counts indicate how often an artist

appears in raw data, giving estimates of rarity or prominence. While these can be interpreted in several ways, random forests handle these varying interpretations well.

Other attempts at gathering outside data, such as web scraping Fuji Arts or Wikipedia, failed due to my lack of experience with scraping (using vibe-coded datascrappers) and internet connectivity issues.

Validation

Given the hundreds of dummy variables created for series, artist, and condition (many infrequent, but often important for reducing error), the max features parameter had to be cross-validated prior to the model itself and was expected to be high. This was done by running the model across 12 different max feature sizes (from 10% to 100%). Each size was tested 5 times to calculate mean and variance. Based on this, I chose 50% as my max feature size (rather than the standard square root of feature count), since many features were infrequent (e.g., specific damage types) while others (e.g., clearance or low starting bid) were critical and shouldn't be missed. Claude also recommended 50% because it lay within the margin of error of the 100% max features prediction.

After this, I performed both 5-fold and 10-fold cross-validations with the new max features setting. Seeing high correlation and low out-of-bag error consistently, I can cautiously estimate that this model is a solid overall predictor.

5 Fold R^2 Mean	0.809
5 Fold RMSE log	0.362
10 Fold R^2 Mean	0.819
10 Fold RMSE log	0.352

Interpretation of the Market

Prices on Fuji Arts are determined by multiple factors, most of which are present in the dataset. Based on prior research and model results, the most important factors include:

Starting Bid and Sales

Fuji Arts starts bids at either an estimated price for the artwork or a low fixed price ($20–$40) during special offers. Many items go unsold and reappear multiple times at slightly reduced starting bids before either selling, moving to clearance, or being removed. For Fuji Arts

specifically, starting bid structure has significant influence, particularly given that lower-valued pieces are sold to a niche hobbyist base. Final bid price is often influenced more by Fuji Arts' appraisal than by buyers.

Artist

Artist is a factor in price, though not overwhelming. Artists from the “golden age of ukiyo-e” make up a vast majority of the highest-valued pieces, but also many low-valued ones. Because woodblock art is easily reproducible, famous artists had their works printed and preserved heavily. As a result, artist-price relationships are not clearly linear; however, when considered alongside other variables, artist becomes a prominent predictor. In the model, artists like Yoshitoshi were consistently evaluated as having high feature importance.

Edition

Edition is the most prominent factor not included in the model. As a woodblock is used, print quality degrades; earlier editions generally command higher prices. Identifying edition is difficult because it is rarely explicitly documented and must be estimated through visual quality inspection. While allowed, estimating edition requires significant time investment to build and train appropriate models. I suspect the large errors in high-value predictions were due to leaving out edition, as exceptional quality prints could not be identified from the available data.

Condition

Condition plays a minor role in price evaluation. General quality descriptions like “nice”, “good”, or “excellent” made little difference (and were omitted), but presence or absence of damage did. Common damage types, such as stains, ranked among major predictors. Splitting damage into specific types rather than a general flag yielded an increase in OOB R^2, confirming minor model improvement. Condition also noted multi-panel prints, which were uncommon but generally indicated higher value.

Series

Pieces often came in series. Popular series sold for higher values, and series prices generally clustered together. For example, pieces in the “One Hundred Roles of Baiko” series clustered between $150 and $190.

Summary

Each factor plays a clear role, but because they act together, they do not correlate simply with price. Nevertheless, pieces with similar condition, series, and artist yield similar prices. Combined with Fuji Arts' specific starting bid mechanisms, market pricing remains complex and multifaceted.

Citation for additional data:

Khan, S. (2021). Ukiyo-e Dataset [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.13120879

Variable name	Description	Year collected
Artist_count	Number of works attributed to the artist in dataset	2021
Era	Broad historical period associated with the artist	2021